Natural Language Processing
Lab 9: Neural Models - LSTM

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4-5 November 2026

Long Short-Term Memory (LSTM)

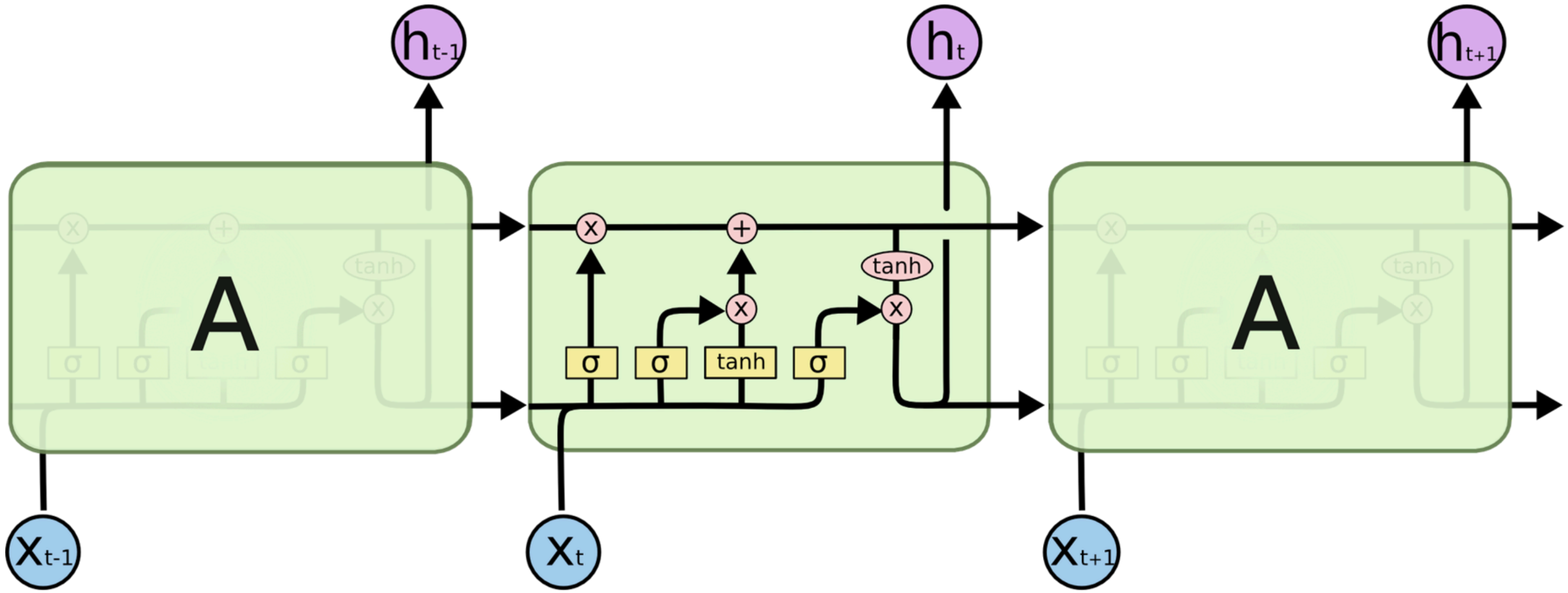


Image: Olah, 2015

Long Short-Term Memory (LSTM)

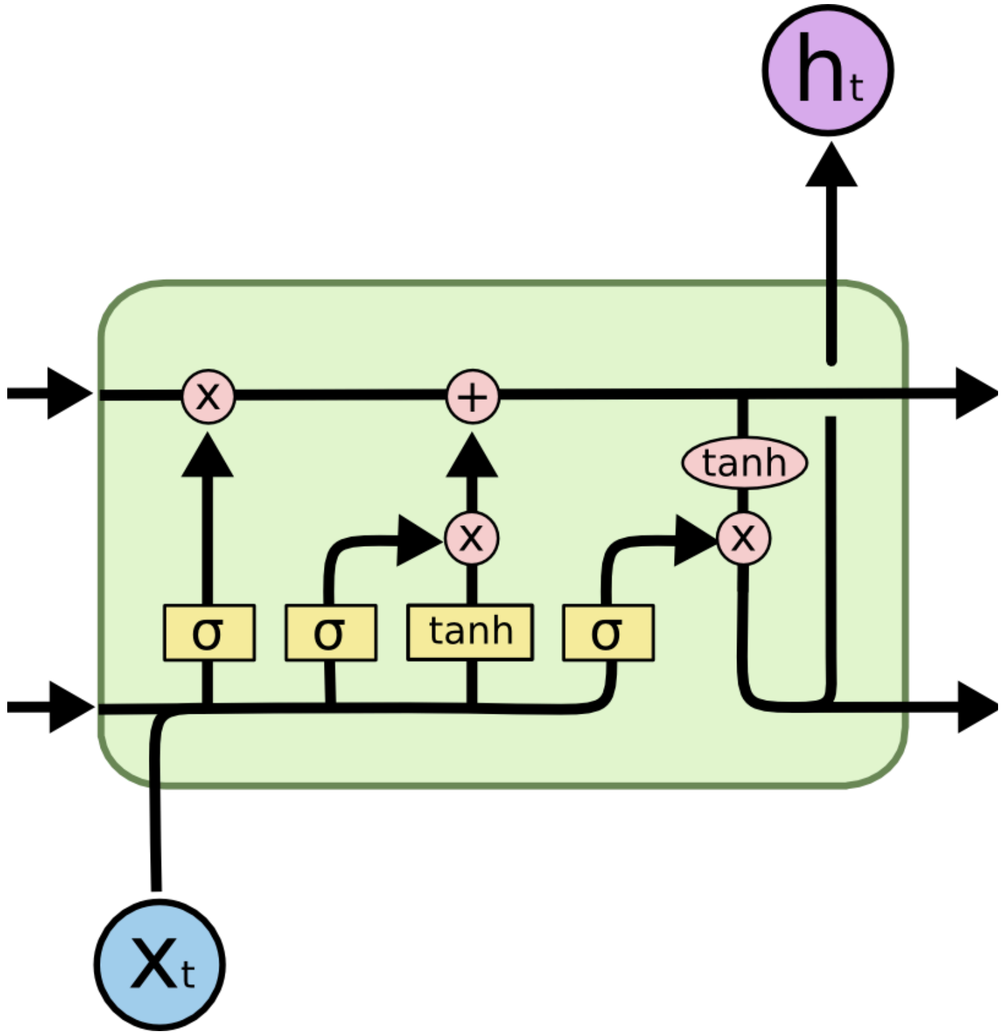


Image: Olah, 2015

Long Short-Term Memory (LSTM)

C_t: cell state (long-term memory)

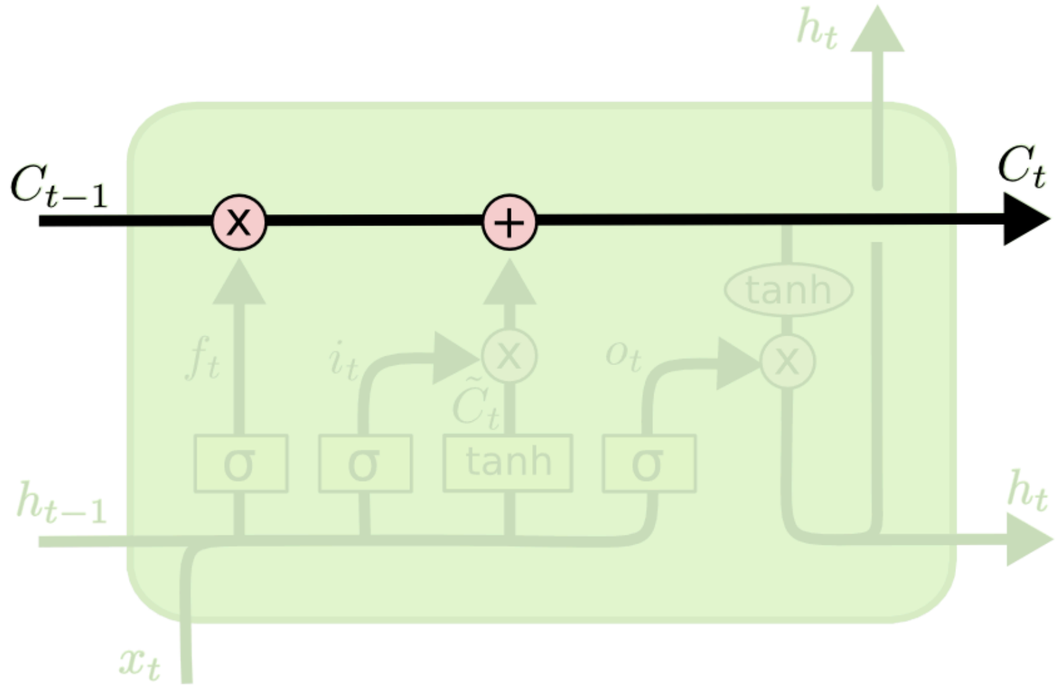


Image: Olah, 2015

Long Short-Term Memory (LSTM)

C_t: cell state (long-term memory)

- hidden_size number of values (e.g,, 64)
- range: (-∞, +∞)
- no activations (linear only)

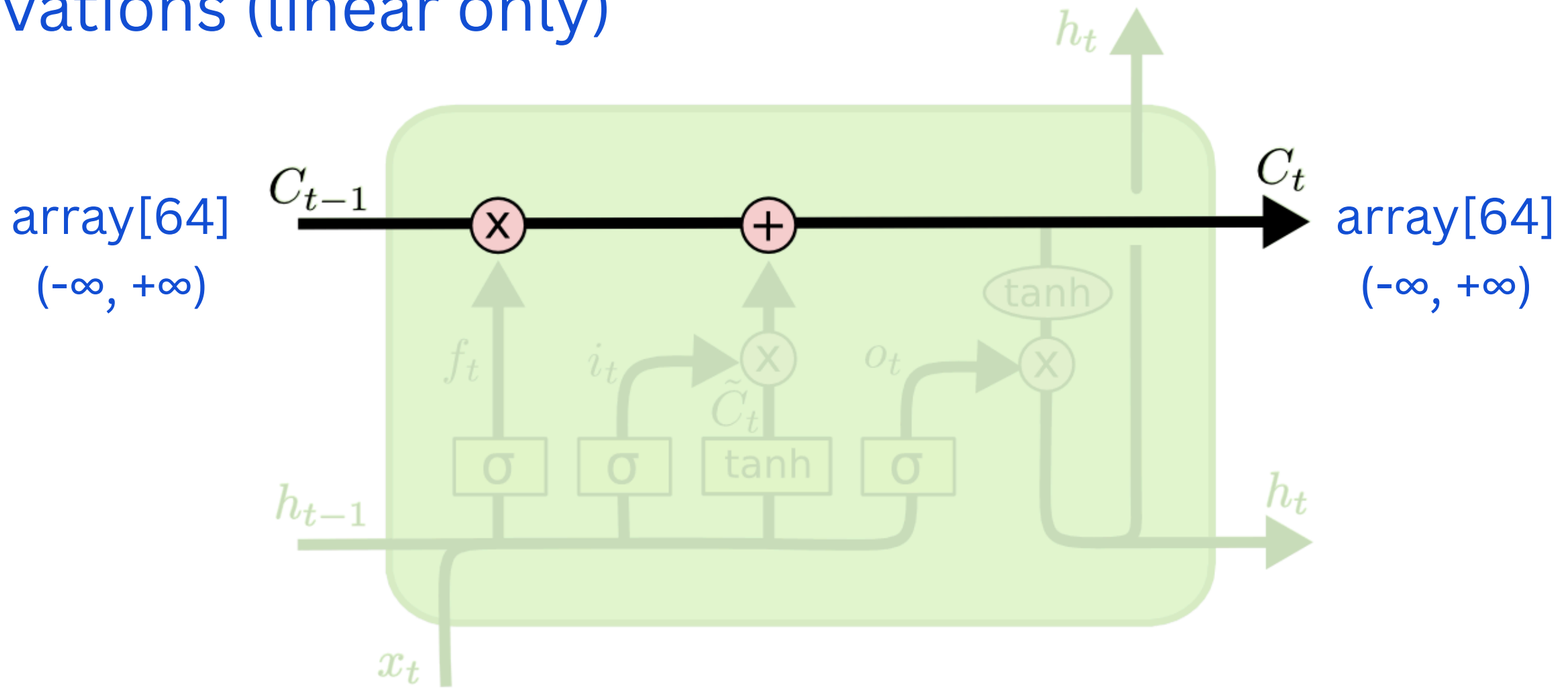
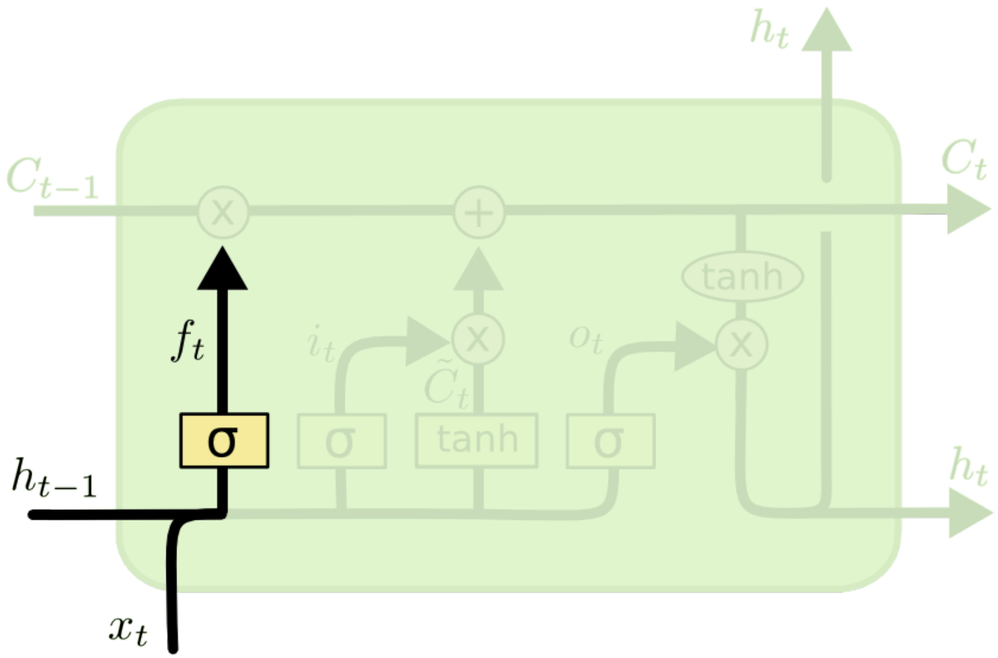


Image: Olah, 2015

Long Short-Term Memory (LSTM)

Step 1: How much to keep/forget per-line of long-term memory?



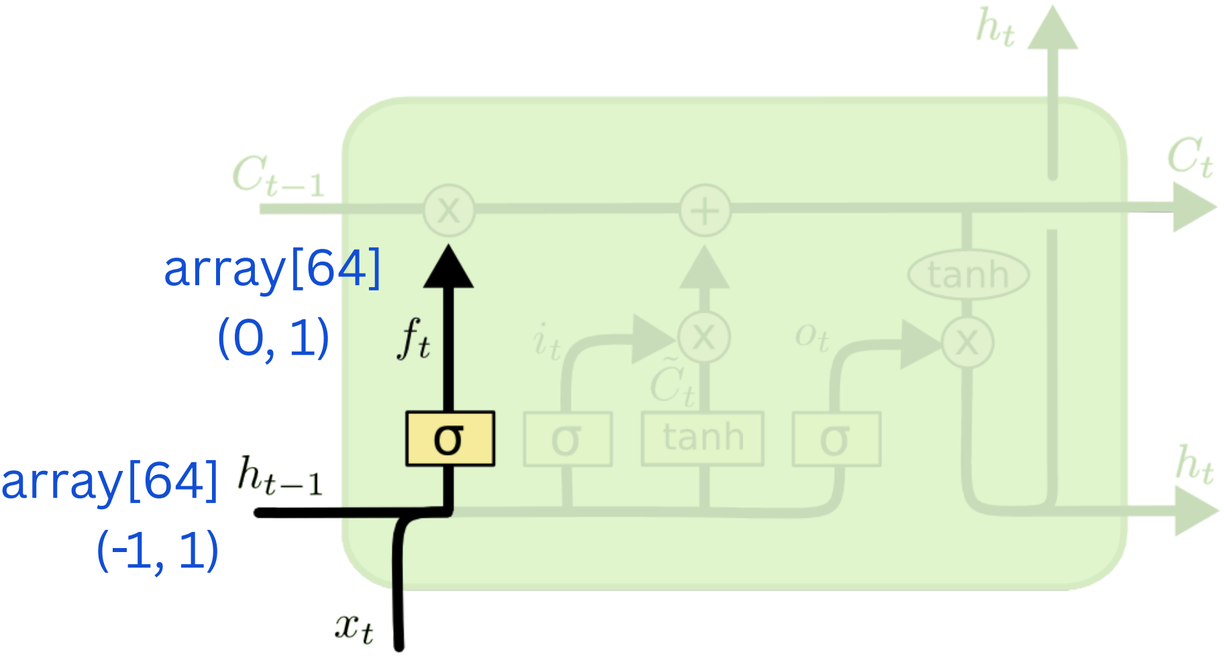
$$f_t = \sigma\left(W_f \cdot [h_{t-1}, x_t] \; + \; b_f\right)$$

Image: Olah, 2015

Long Short-Term Memory (LSTM)

Step 1: How much to keep/forget per-line of long-term memory?

- Last hidden state (h_t-1) and current input (x_t) determine how much to keep/forget
- f_t range is (0, 1) from sigmoid output



array[300] ← text: embedding dimensions

sigmoid(...) → (0, 1)

$$f_t = \sigma\left(W_f \cdot [h_{t-1}, x_t] \; + \; b_f\right)$$

array[64, 64+300]

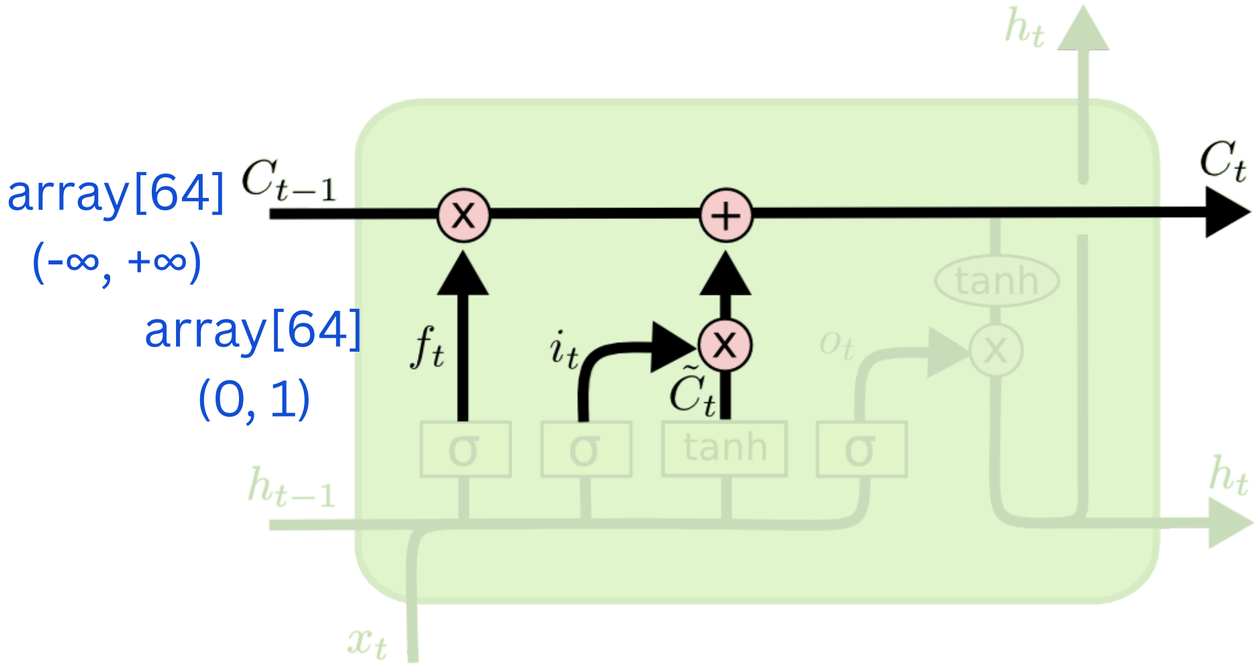
array[64]

Image: Olah, 2015

Long Short-Term Memory (LSTM)

Step 1: How much to keep/forget per-line of long-term memory?

- The previous cell state (C_t-1) is “pointwise multiplied” by f_t
- Mostly keep if f_t ~1
- Mostly forget if f_t ~0

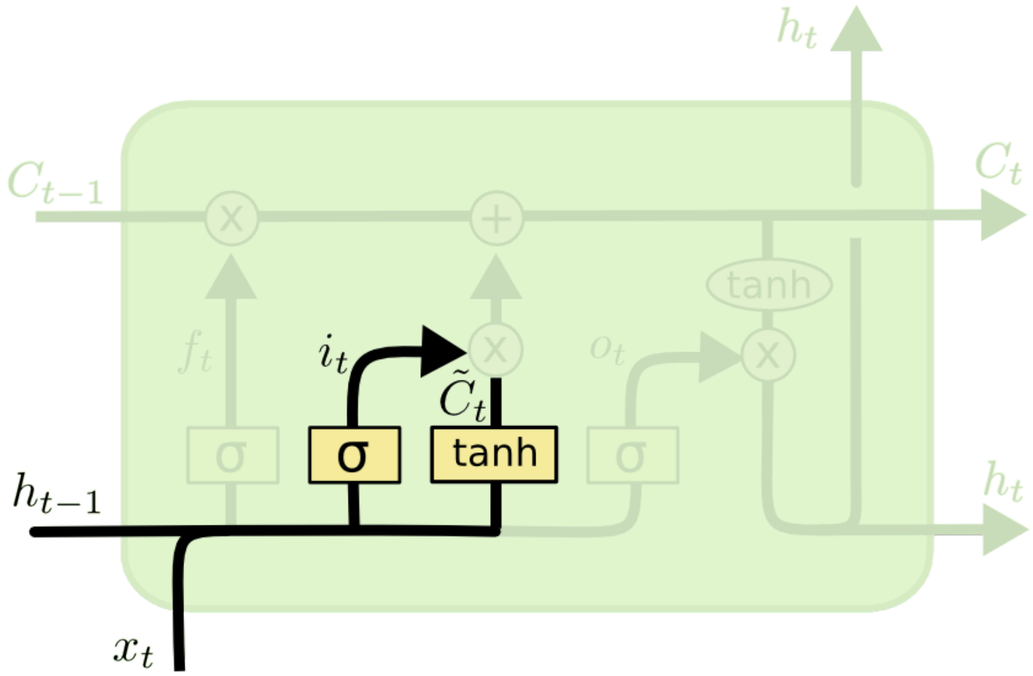


$$C_t = f_t * C_{t-1} + \ldots$$

Image: Olah, 2015

Long Short-Term Memory (LSTM)

Step 2: What new information and how much of it to ADD to C_t?



$$i_t = \sigma\left(W_i \cdot [h_{t-1}, x_t] \; + \; b_i\right)$$

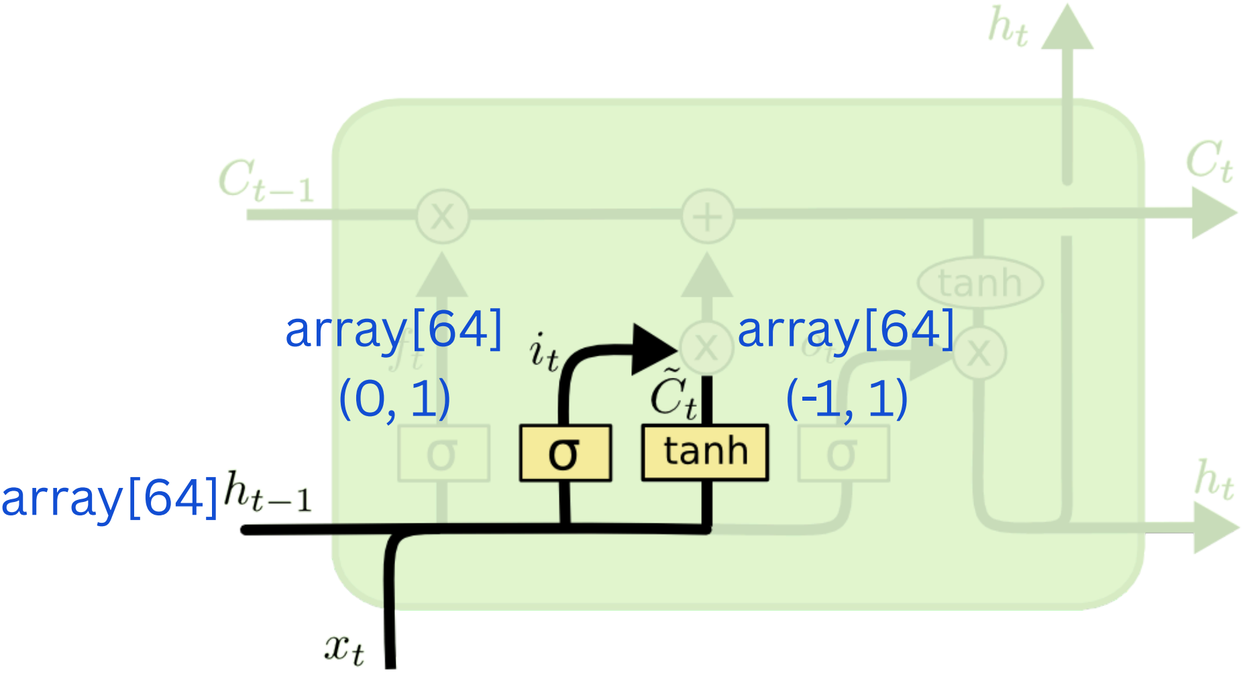
$$\tilde{C}_t = \tanh(W_C \cdot [h_{t-1}, x_t] \; + \; b_C)$$

Image: Olah, 2015

Long Short-Term Memory (LSTM)

Step 2: What new information and how much of it to ADD to C_t?

- Cand_t is the new information (range of (-1, 1) from tanh)
- i_t is how much to use (range of (0, 1) from sigmoid)



array[300] ← text: embedding dimensions

sigmoid(...) → (0, 1)

tanh(...) → (-1, 1)

$$i_t = \sigma\left(W_i \cdot [h_{t-1}, x_t] \; + \; b_i\right)$$

$$\tilde{C}_t = \tanh\left(W_C \cdot [h_{t-1}, x_t] \; + \; b_C\right)$$

array[64, 64+300]

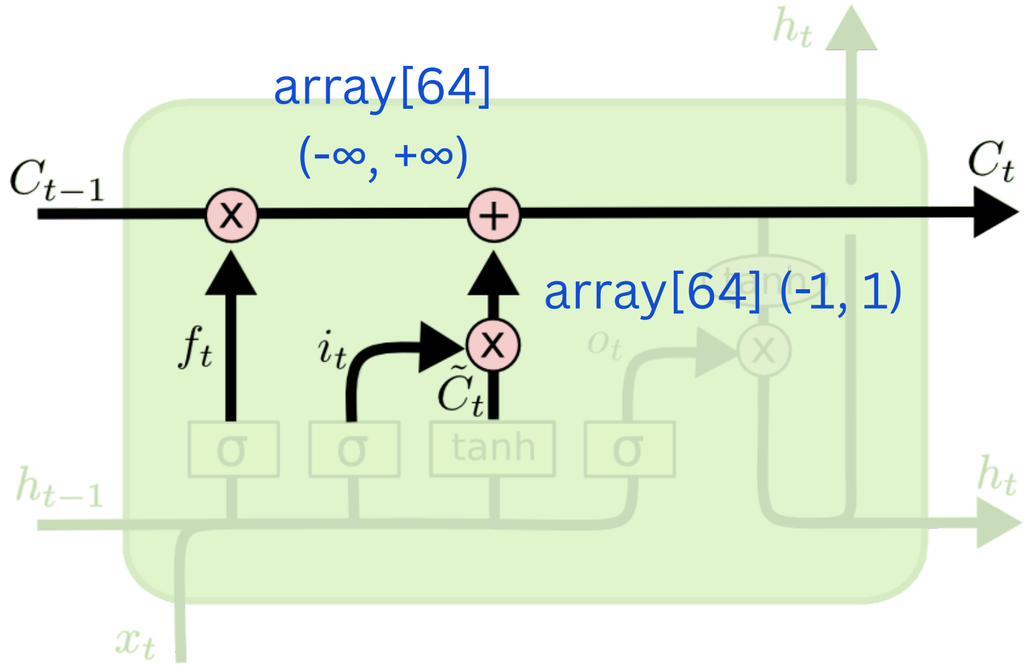
array[64]

Image: Olah, 2015

Long Short-Term Memory (LSTM)

Step 2: What new information and how much of it to ADD to C_t?

- The output of these two activations are “pointwise multiplied”
- Then, that result is “pointwise added” to the “remembered” cell state
- The last result is the final C_t of this time step

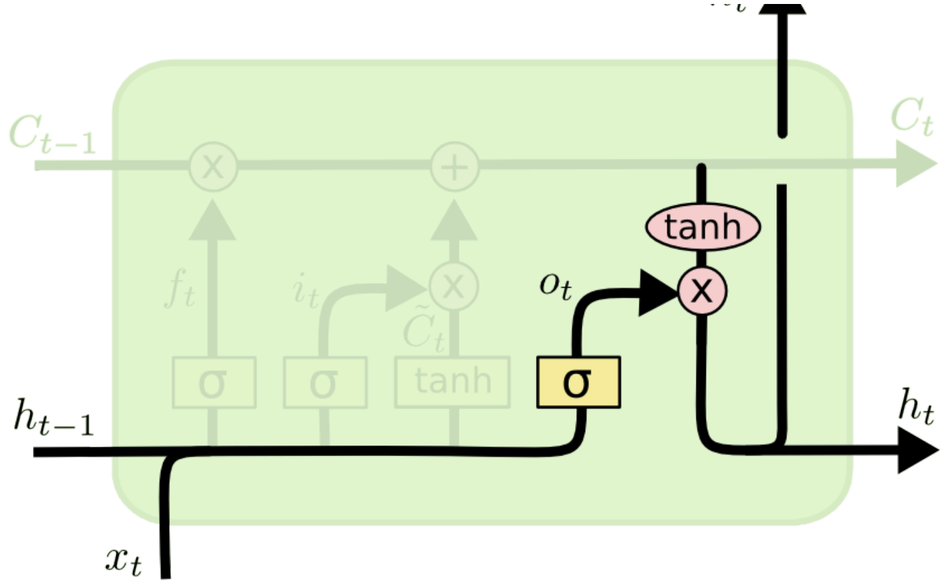


$$C_t = f_t * C_{t-1} + i_t * \tilde{C}_t$$

Image: Olah, 2015

Long Short-Term Memory (LSTM)

Step 3: Determine the next hidden state (short-term memory)



$$o_t = \sigma\left(W_o\left[h_{t-1}, x_t\right] + b_o\right)$$

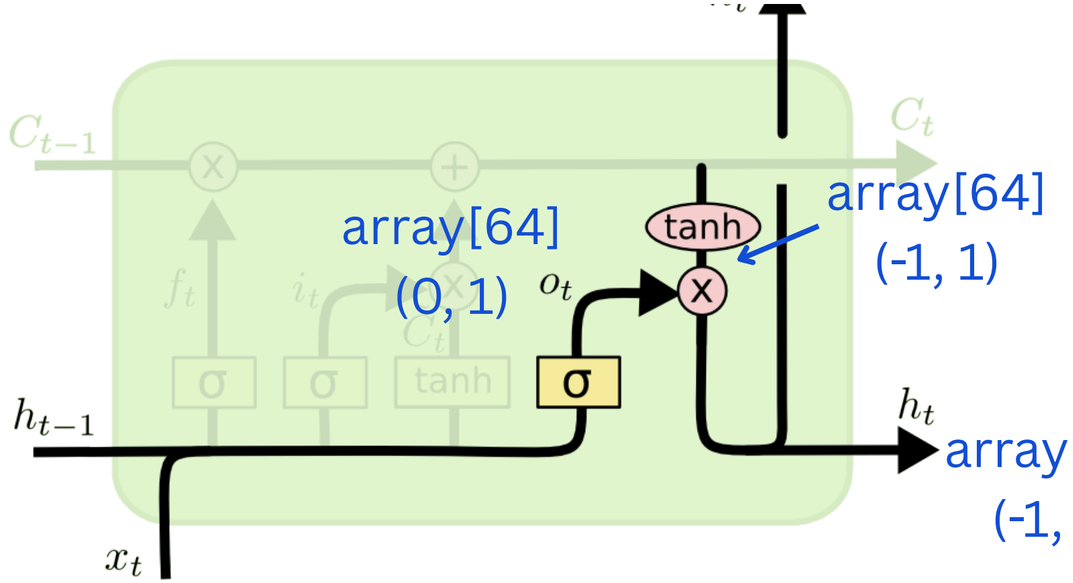
$$h_t = o_t * \tanh\left(C_t\right)$$

Image: Olah, 2015

Long Short-Term Memory (LSTM)

Step 3: Determine the next hidden state (short-term memory)

- The last hidden state, input, and current cell state are all used to determine the next hidden state
- The hidden state stays in the range of (-1, 1) thus avoiding the vanishing/exploding gradient problem!



array[64, 64+300]

$$o_t = \sigma\left(W_o\left[h_{t-1}, x_t\right] + b_o\right)$$

$$h_t = o_t * \tanh\left(C_t\right)$$

array[64]

Image: Olah, 2015

Long Short-Term Memory (LSTM)

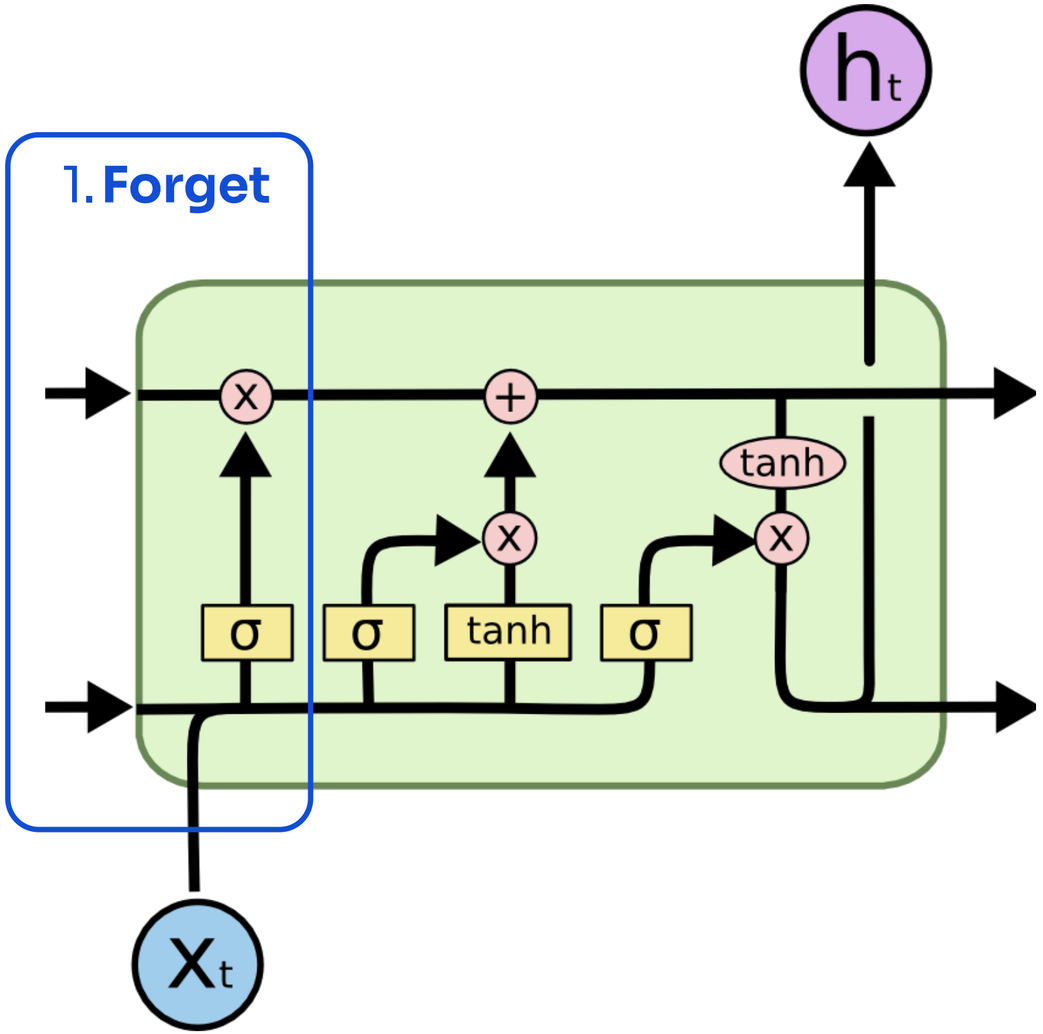


Image: Olah, 2015

Long Short-Term Memory (LSTM)

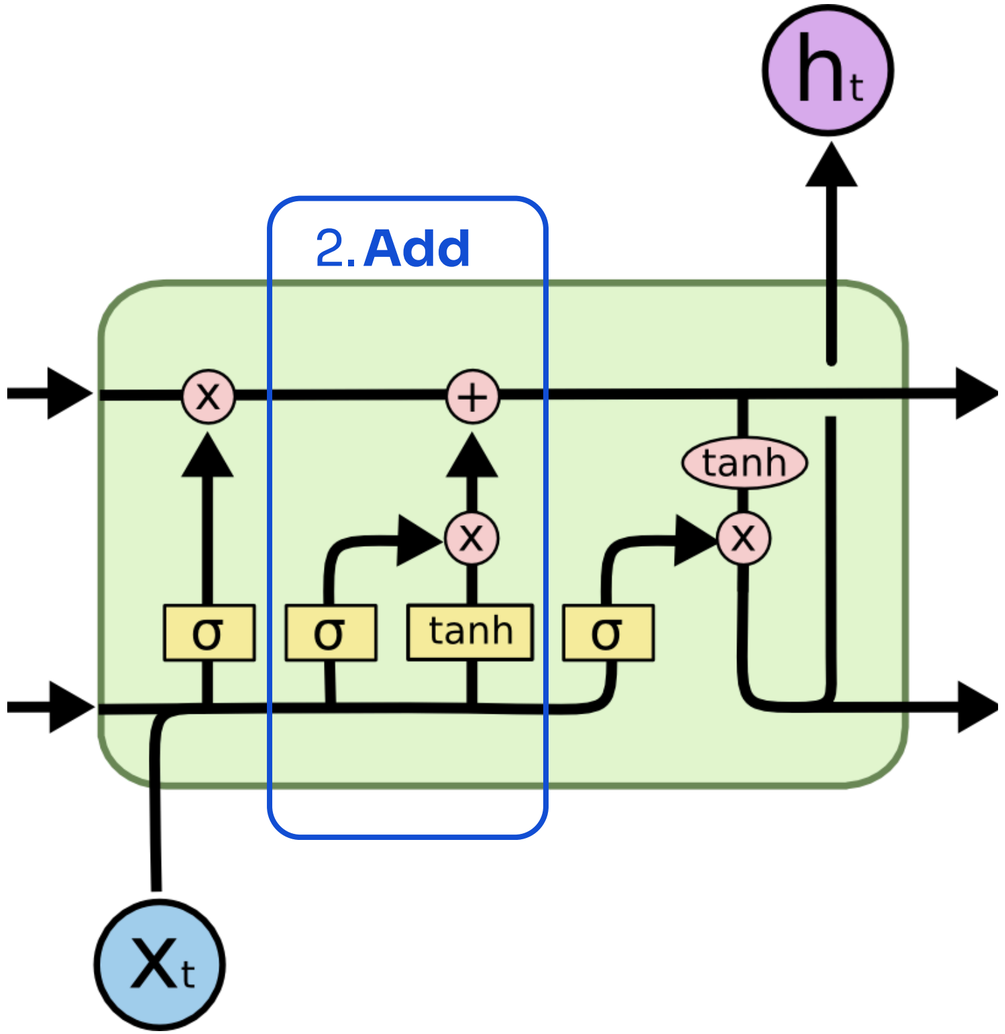


Image: Olah, 2015

Long Short-Term Memory (LSTM)

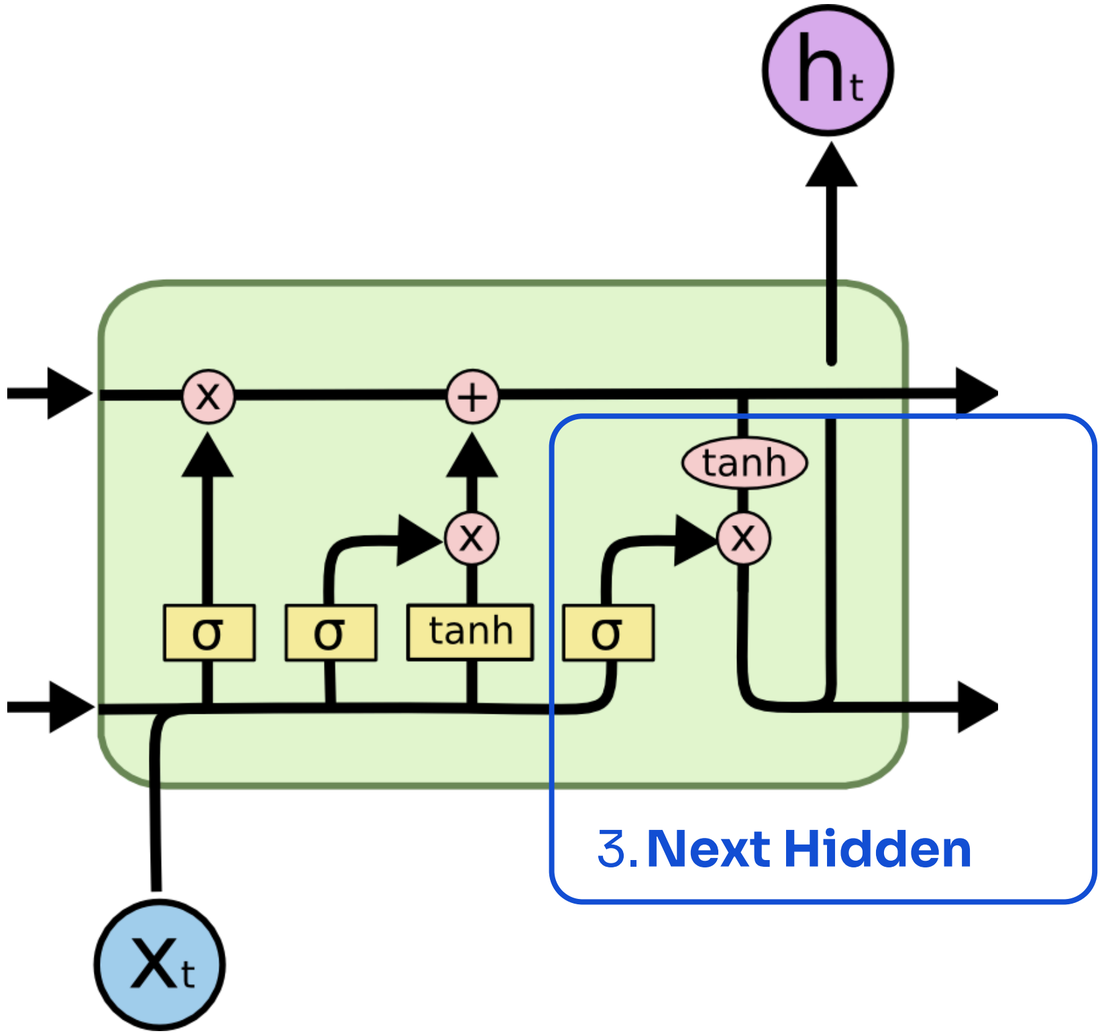


Image: Olah, 2015

Language Modeling: Next Char/Word Pred.

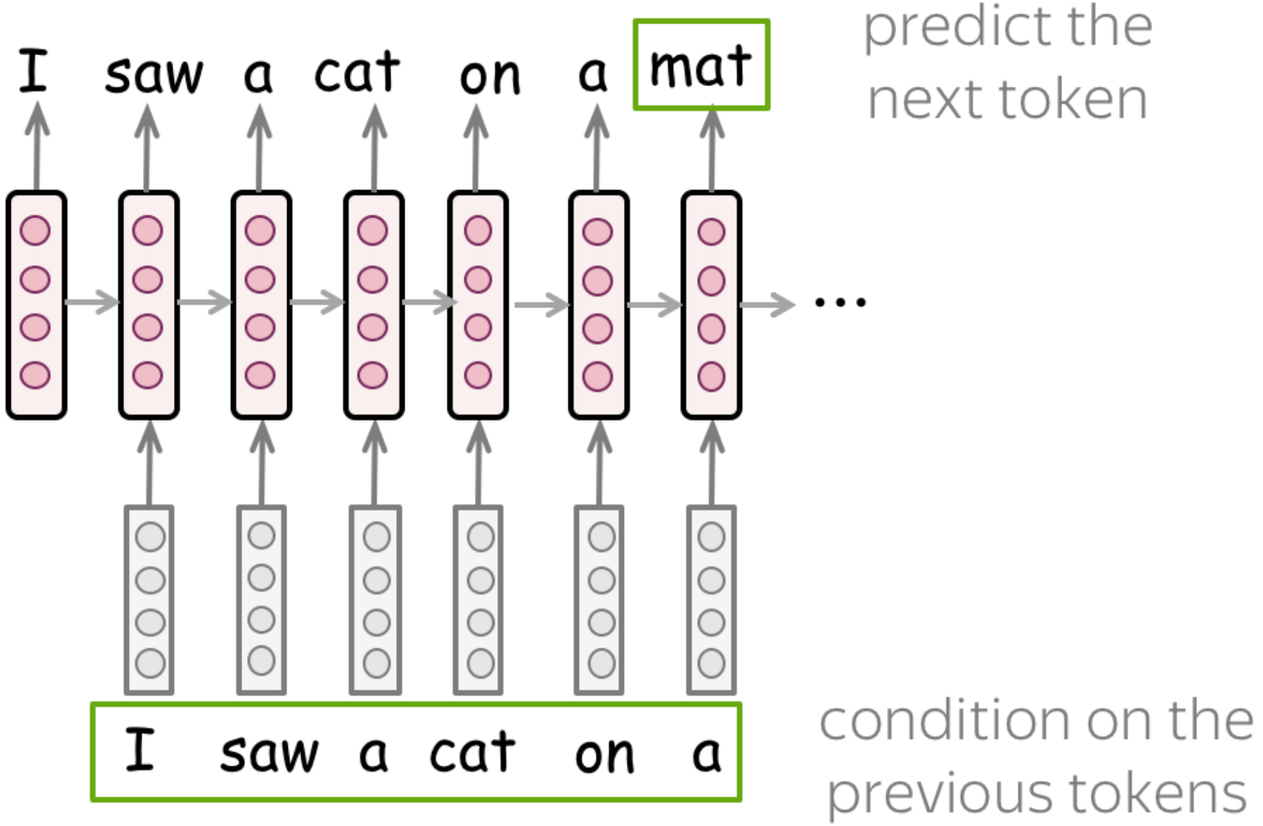


Image: Voita, 2026

Language Modeling: Translation (seq2seq)

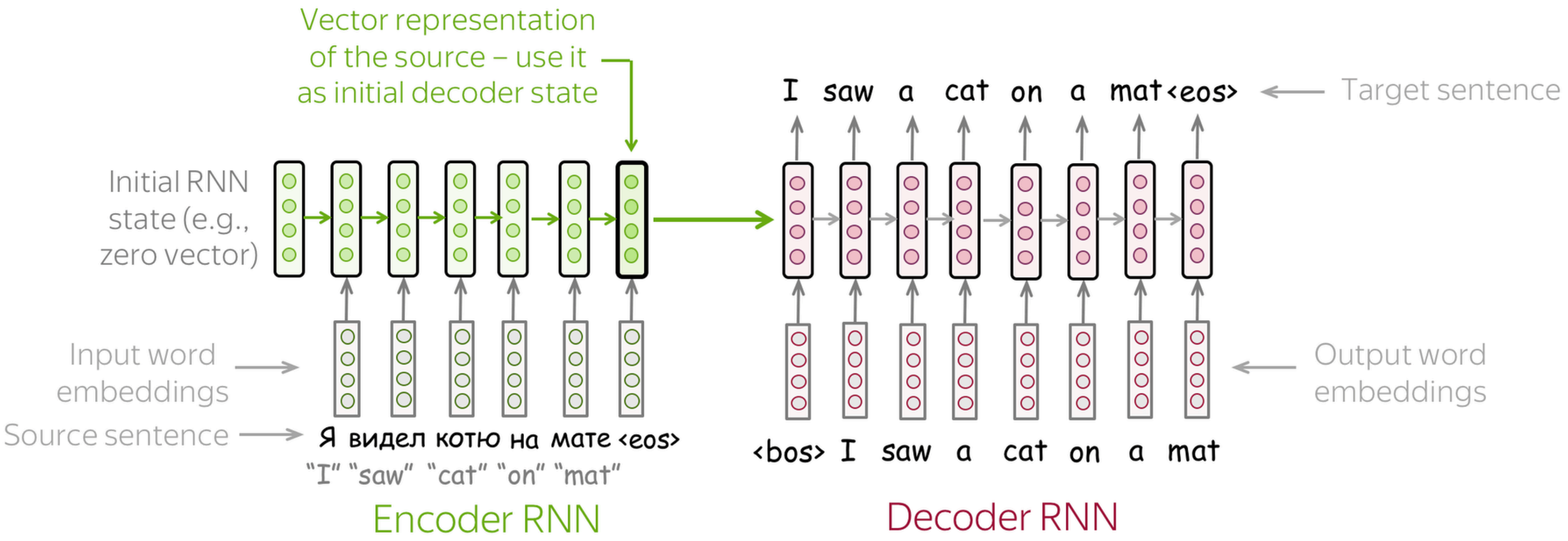


Image: Voita, 2026

Next Week: Attention and Transformers!

